

Homework A2 | ECON 0100 | Fall 2026

Due: Sunday, Sep. 6

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely, and show your work so you can later understand your thought process. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

The Hogsmeade Candy Shop Saga

Honeydukes and Zonkos are two candy shops in Hogsmeade. Both shops make two types of popular candy, Chocolate Frogs (F) and Ice Mice (M). Devoting all their resources to one good, Honeydukes can make 100 pounds of F or 75 pounds of M each week, and Zonkos can make 200 pounds of F or 100 pounds of M .

Construct a production table and an opportunity cost table to keep track of your work as you move through the questions.

Q1 | Absolute Advantage

- a) Which shop has the absolute advantage in Chocolate Frogs?
- b) Which shop has the absolute advantage in Ice Mice?

Q2 | Opportunity Costs

In Homework A1 you found Honeydukes' opportunity cost of 1F. Fill in the rest of the opportunity cost table.

- a) What is Honeydukes' opportunity cost of 1M?
- b) What is Zonkos' opportunity cost of 1F?

c) What is Zonkos' opportunity cost of 1M?

Q3 | Comparative Advantage

a) Which shop has the comparative advantage in Chocolate Frogs?

b) Which shop has the comparative advantage in Ice Mice?

c) Zonkos has the absolute advantage in both goods. Explain why it can't have the comparative advantage in both.

Q4 | Changes in Labor and Technology

Gringotts decides to fund Zonkos' purchase of a more capable Ice Mice machine, doubling Zonkos' Ice Mice production. At the same time, Honeydukes triples their labor hours, tripling their productive capacity. Construct an updated production table and opportunity cost table before answering.

a) After both changes, which shop has the absolute advantage in each good?

b) After both changes, what is Honeydukes' opportunity cost of 1F? What is Zonkos'?

c) After both changes, which shop has the comparative advantage in each good?

<!-- The questions are free response on paper; students submit selections on Gradescope. The selection form and answer key live in Homework_A2_gradescope.md. ANSWER KEY (not for students) Production table: Honeydukes 100F or 75M; Zonkos 200F or 100M. OC table: Honeydukes $OC(1F) = 3/4 M$, $OC(1M) = 4/3 F$; Zonkos $OC(1F) = 1/2 M$, $OC(1M) = 2 F$. Q1a: Zonkos Q1b: Zonkos Q2a: $4/3$ pounds of F Q2b: $1/2$ pound of M Q2c: 2 pounds of F Q3a: Zonkos ($1/2 < 3/4$) Q3b: Honeydukes ($4/3 < 2$) Q3c: opportunity costs are reciprocals, so a lower OC in one good means a higher OC in the other; whoever gives up less F per M must give up more M per F After the changes: Honeydukes 300F or 225M; Zonkos 200F or 200M. Q4a: Honeydukes in

both goods Q4b: Honeydukes $3/4$ pound of M (unchanged; tripling labor shifts the PPF without changing its slope); Zonkos 1 pound of M Q4c: Honeydukes in Chocolate Frogs ($3/4 < 1$), Zonkos in Ice Mice ($1 < 4/3$) $\rightarrow$